

Corolario 1. Sea $f \in \mathcal{H}(\mathbb{D})$, entonces $\sup_{z \in \mathbb{D}} f^{\natural}(z) = \sup_{T \in \text{Aut}(\mathbb{D})} (f \circ T)^{\natural}(0)$.

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Demostración: Por el lema-invarianza-derivada-hiperbolica-aut-disco-unidad/lema 1, tenemos que $\forall T \in \text{Aut}(\mathbb{D}) : (f \circ T)^{\natural}(0) = f^{\natural}(T(0))$.

$\boxed{\leq}$ Sea $z \in \mathbb{D}$. Entonces, $\tau_z \in \text{Aut}(\mathbb{D})$ satisface que $\tau_z(0) = z$, luego

$$f^{\natural}(z) = f^{\natural}(\tau_z(0)) = (f \circ \tau_z)^{\natural}(0) \leq \sup_{T \in \text{Aut}(\mathbb{D})} (f \circ T)^{\natural}(0).$$

Como z era arbitrario, $\sup_{z \in \mathbb{D}} f^{\natural}(z) \leq \sup_{T \in \text{Aut}(\mathbb{D})} (f \circ T)^{\natural}(0)$.

$\boxed{\geq}$ Sea $T \in \text{Aut}(\mathbb{D})$. Entonces, $T(0) \in \mathbb{D}$, luego $(f \circ T)^{\natural}(0) = f^{\natural}(T(0)) \leq \sup_{z \in \mathbb{D}} f^{\natural}(z)$ y, como T era arbitrario, $\sup_{T \in \text{Aut}(\mathbb{D})} (f \circ T)^{\natural}(0) \leq \sup_{z \in \mathbb{D}} f^{\natural}(z)$. ■